

Program 10: Demonstrate kubernetes dashboard and cli for cluster monitoring

File structure:

deployment.yaml

services.yaml

deployment.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

name: nginx-deployment

labels:

app: nginx

spec:

replicas: 2

selector:

matchLabels:

app: nginx

template:

metadata:

labels:

app: nginx

spec:

containers:

- name: nginx

image: nginx:latest

ports:

- containerPort: 80

services.yaml

apiVersion: v1

kind: Service

metadata:

name: nginx-service

spec:

selector:

app: nginx

ports:

- protocol: TCP

port: 80

targetPort: 80

type: NodePort

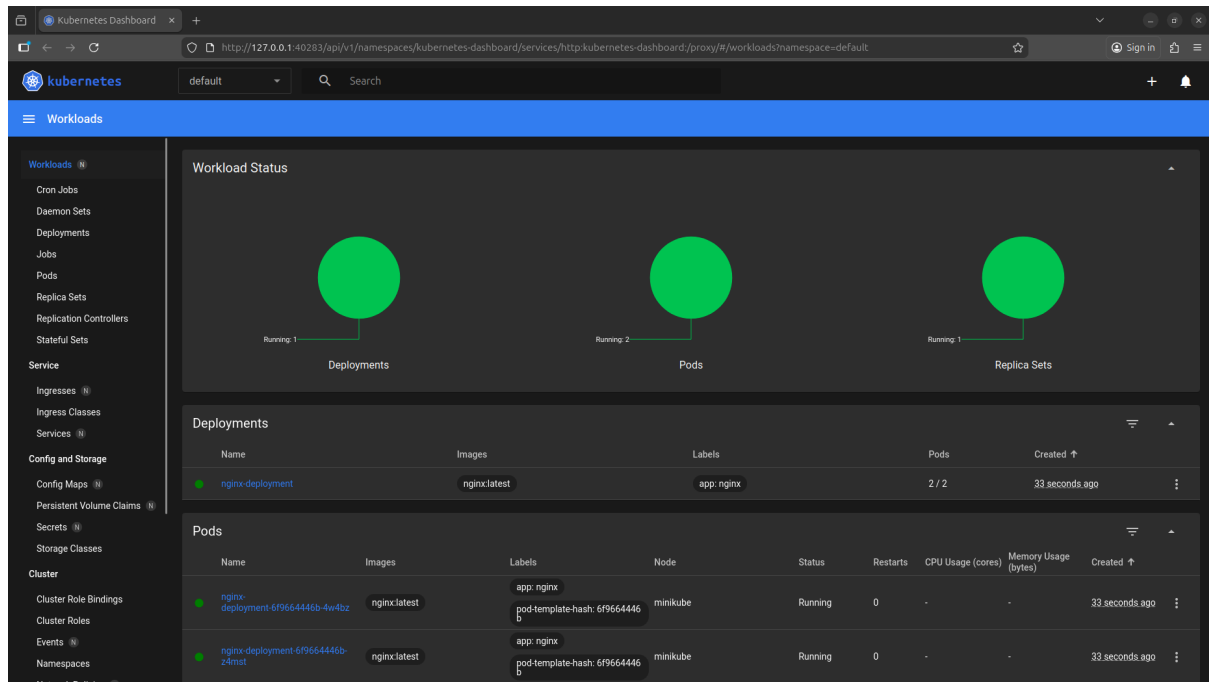
Start minikube with assigning 2 cpus and 4096 byte memory

#minikube start --cpus=2 --memory=4096

Check the status to verify all the services are running

#minikube status

Open the kubernetes dashboard
#minikube dashboard



Create the pod from YAML files
#kubectl apply -f deployment.yaml
#kubectl apply -f services.yaml

List all the resources
#kubectl get pods
#kubectl get svc
#kubectl get deployment

Log the pod
#kubectl logs nginx-deployment-6f9664446b-4w4bz

```
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/lab10
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/lab10
deployment.apps/nginx-deployment created
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab10$ kubectl apply -f services.yaml
service/nginx-service created
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-4w4bz   1/1     Running   0           11s
nginx-deployment-6f9664446b-z4n8t   1/1     Running   0           11s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab10$ kubectl get svc
NAME      TYPE      CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes  ClusterIP  10.96.0.1     <none>        443/TCP    9h
nginx-service  NodePort   10.111.28.57  <none>        80:30937/TCP  11s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab10$ kubectl get deployments.apps
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  2/2     2             2           21s
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab10$ kubectl logs nginx-deployment-6f9664446b-4w4bz
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/19 04:17:09 [notice] #1: using the "epoll" event method
2025/12/19 04:17:09 [notice] #1: nginx/1.29.4
2025/12/19 04:17:09 [notice] #1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/19 04:17:09 [notice] #1: OS: Linux 6.14.0-36-generic
2025/12/19 04:17:09 [notice] #1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/19 04:17:09 [notice] #1: start worker processes
2025/12/19 04:17:09 [notice] #1: start worker process 29
2025/12/19 04:17:09 [notice] #1: start worker process 30
2025/12/19 04:17:09 [notice] #1: start worker process 31
2025/12/19 04:17:09 [notice] #1: start worker process 32
2025/12/19 04:17:09 [notice] #1: start worker process 33
2025/12/19 04:17:09 [notice] #1: start worker process 34
2025/12/19 04:17:09 [notice] #1: start worker process 35
2025/12/19 04:17:09 [notice] #1: start worker process 36
2025/12/19 04:17:09 [notice] #1: start worker process 37
2025/12/19 04:17:09 [notice] #1: start worker process 38
2025/12/19 04:17:09 [notice] #1: start worker process 39
2025/12/19 04:17:09 [notice] #1: start worker process 40
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab10$
```

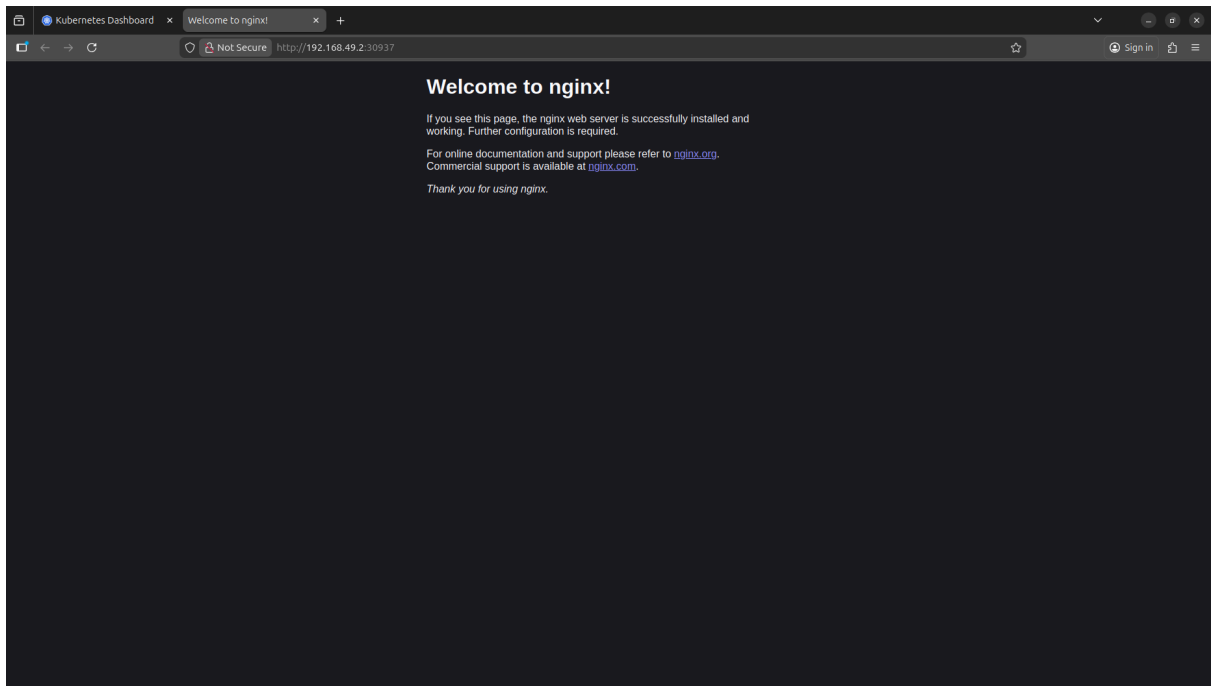
Describe the pod

#kubectl describe pod nginx-deployment-6f9664446b-4w4bz

```
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/lab10
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ: ~/Desktop/LabPrograms/lab10
2025/12/19 04:17:09 [notice] #1: start worker process 39
2025/12/19 04:17:09 [notice] #1: start worker process 40
1rv24mc016_anantha_m@anantha-ASUS-TUF-Gaming-A15-FA506IHRB-FA506IHRZ:~/Desktop/LabPrograms/lab10$ kubectl describe pod nginx-deployment-6f9664446b-4w4bz
Name:                                nginx-deployment-6f9664446b-4w4bz
Namespace:                           default
Priority:                              0
Service Account:                       default
Node:                                 minikube/192.168.49.2
Start Time:                           Fri, 19 Dec 2025 09:47:02 +0530
Labels:                                app=nginx
                                         pod-template-hash=6f9664446b
Annotations:                           <none>
Status:                                Running
IP:                                    10.244.0.61
IPs:                                   <none>
IP:                                    10.244.0.61
Controlled By:                         ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:   docker://d1570e84f55105b0fa4e3212b32c0e47db2b84c44ae417a11e471ac7e8ccb62a
    Image:          nginx:latest
    Image ID:       docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d18fccc66d19
    Port:          80/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Fri, 19 Dec 2025 09:47:09 +0530
    Ready:        True
    Restart Count: 0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-5nj44 (ro)
Conditions:
  Type                 Status
  PodReadyToStartContainers  True
  Initialized           True
  Ready                 True
  ContainersReady       True
  PodScheduled          True
Volumes:
  kube-api-access-5nj44:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:       kube-root-ca.crt
    ConfigMapOptional:    <not specified>
```

Start the nginx service using minikube

#minikube service nginx-service



Deleting resources

```
#kubectl delete pods --all  
#kubectl delete svc--all  
#kubectl delete deployments --all
```